Name: Sheikh Wasimuddin
Branch: CSE-IOT
Batch: A-4
Roll No. :IoT_63

MSPA :-1

PROBLEM STATEMENT 1:

Create using java collection the system should allow adding food item to an order, removing item and calculating the total bill use to a store order item and to store the price of each food item.

CODE:

```
//mspa1_sheikh wasimuddin_iot63
import java.util.*;
class Item {
   String name;
   double price;

   Item(String n, double p) {
      name = n;
      price = p;
   }
}
public class MSPA1PS1 {
   static Scanner sc = new Scanner(System.in);
   static ArrayList<Item> list = new ArrayList<>();
   public static void addItem() {
      System.out.print("Enter food name: ");
      String name = sc.nextLine();
      System.out.print("Enter price: ");
      double price = sc.nextDouble();
      sc.nextLine();
      list.add(new Item(name, price));
   }
   public static void removeItem() {
      System.out.print("Enter item name to remove: ");
      String name = sc.nextLine();
      boolean found = false;
      for (int i = 0; i < list.size(); i++) {
         if (list.get(i).name.equalsIgnoreCase(name)) {
            list.remove(i);
            found = true;
            break;
         }
      }
      if (!found) {
         System.out.println("Item not found");
```

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```
        }
    }
    public static void viewOrder() {
        if (list.isEmpty()) {
            System.out.println("Order is empty");
            return;
        }
        for (Item i : list) {
            System.out.println(i.name + " - " + i.price);
        }
    }
    public static void totalPrice() {
        double total = 0;
        for (Item i : list) {
            total += i.price;
        }
        System.out.println("Total Price: " + total);
    }
    public static void main(String[] args) {
        int choice;
        do {
            System.out.println("\n1 Add Item");
            System.out.println("2 Remove Item");
            System.out.println("3 View Order");
            System.out.println("4 Total Price");
            System.out.println("5 Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    addItem();
                    break;
                case 2:
                    removeItem();
                    break;
                case 3:
                    viewOrder();
                    break;
                case 4:
                    totalPrice();
                    break;
            }
        } while (choice != 5);
    }
```

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}

OUTPUT:

```
MSPA1PS1.java
MSPA1PS1.java > ...
1   //problem statement 1: create using java collection the system should allow adding food items to an order,removing item
2
3   //mspa1_sheikh wasimuddin_iot63
4   import java.util.*;

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\student\Desktop>  & 'C:\Program Files\Java\jdk-22\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' '
C:\Users\student\AppData\Roaming\Code\User\workspaceStorage\7e71a37d41ebfa530b4a4bcfc06abf27\redhat.java\jdt_ws\Desktop_76766
649\bin' 'MSPA1PS1'

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 1
Enter food name: BURGER
Enter price: 149

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 1
Enter food name: PIZZA
Enter price: 399

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
```

```
MSPA1PS1.java
MSPA1PS1.java > ...
1   //problem statement 1: create using java collection the system should allow adding food items to an order,removing item
2
3   //mspa1_sheikh wasimuddin_iot63
4   import java.util.*;

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 1
Enter food name: JUICE
Enter price: 199

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 3
BURGER - 149.0
PIZZA - 399.0
JUICE - 199.0

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 4
Total Price: 747.0
```

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MSPA1PS1.java

```java
//problem statement 1: create using java collection the system should allow adding food items to an order,removing item

//mspa1_sheikh wasimuddin_iot63
import java.util.*;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 2
Enter item name to remove: JUICE

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 3
BURGER - 149.0
PIZZA - 399.0

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 4
Total Price: 548.0
```

MSPA1PS1.java

```java
//problem statement 1: create using java collection the system should allow adding food items to an order,removing item

//mspa1_sheikh wasimuddin_iot63
import java.util.*;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter choice: 2
Enter item name to remove: JUICE

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 3
BURGER - 149.0
PIZZA - 399.0

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 4
Total Price: 548.0

1 Add Item
2 Remove Item
3 View Order
4 Total Price
5 Exit
Enter choice: 5
PS C:\Users\student\Desktop>
```

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PROBLEM STATEMENT 2:

develop a java program to store in a file the program should be reading the file content.

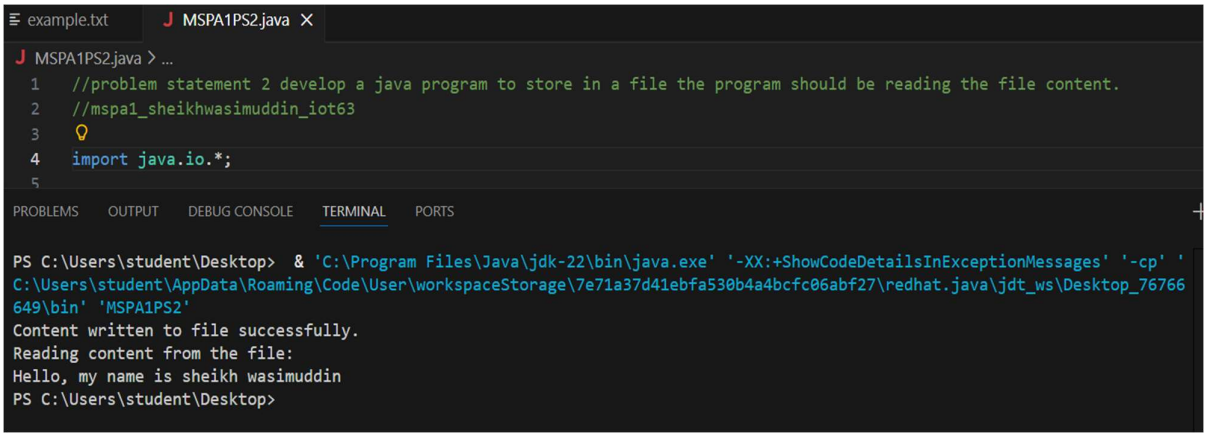
CODE:

```
//mspa1_sheikhwasimuddin_iot63
import java.io.*;
public class MSPA1PS2{
    public static void main(String[] args) {
        String fileName = "example.txt";
        String content = "Hello, my name is sheikh wasimuddin";

        try (FileWriter writer = new FileWriter(fileName)) {
            writer.write(content);
            System.out.println("Content written to file successfully.");
        } catch (IOException e) {
            System.out.println("An error occurred while writing to the file");
            e.printStackTrace();
        }

        try (BufferedReader reader = new BufferedReader(new FileReader(fileName))) {
            String line;
            System.out.println("Reading content from the file:");
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
        } catch (IOException e) {
            System.out.println("An error occurred while reading the file.");
            e.printStackTrace();
        }
    }
}
```

OUTPUT:



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example.txt × MSPA1PS2.java

example.txt

```
1   Hello, my name is sheikh wasimuddin
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\student\Desktop>  & 'C:\Program Files\Java\jdk-22\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' '
C:\Users\student\AppData\Roaming\Code\User\workspaceStorage\7e71a37d41ebfa530b4a4bcfc06abf27\redhat.java\jdt_ws\Desktop_76766
649\bin' 'MSPA1PS2'
Content written to file successfully.
Reading content from the file:
Hello, my name is sheikh wasimuddin
PS C:\Users\student\Desktop>
```